Temporal Wave Theory, Solar Temporal Flux, and Temporal Dilation

Author: Viorel-Mihai Popescu Affiliation: Polytechnic University of Bucharest — TauTime Date: 9 June 2026

1. Formulation of the Partial Differential Equation

The temporal wave probability P(r,t) of the TauTime theory satisfies the one-dimensional spherical wave equation. It describes the propagation of time quanta in the solar gravitational field, at a distance r from the center of the Sun and at moment t:

$$\partial^2 P(r,t)/\partial t^2 - c^2 \cdot \partial^2 P(r,t)/\partial r^2 = 0 \quad (1)$$

where $c = 2.998 \times 10^8$ m/s is the speed of light in vacuum.

1.1 Boundary Conditions

Cancellation condition at the Schwarzschild-related radius (threshold of the solar black hole, 3.8 $R_s^R\odot$):

$$P(r_0, 0) = 0, \quad r_0 = 11{,}222.3 \text{ m} \quad (2)$$

Cancellation condition at the TauTime cosmological radius:

$$P(R, 0) = 0, \quad R = 4.2355 \times 10^{22} \text{ m} \quad (3)$$

These parameters are fundamental in the TauTime structure: r_0 marks the inner limit of the solar time field (black-hole threshold), and R is the cosmological cohesion radius at which the time field cancels out.

2. General Solution — Degree 2 in r, Degree 2 in t

A solution of degree 2 in r and degree 2 in t is proposed, satisfying both boundary conditions by construction:

$$P(r,t) = A \cdot [\, (r - r_0)(r - R) + \varepsilon \cdot t + c^2 \cdot t^2 \,] \quad (4)$$

2.1 Verifying the Solution in the PDE

$$\partial^2 P/\partial t^2 = 2\, A\, c^2 \quad (5)$$

$$\partial^2 P/\partial r^2 = 2\, A \quad (6)$$

Substituting into equation (1):

$$2Ac^2 - c^2 \cdot 2A = 0 \quad \checkmark \quad (7)$$

Conclusion: Solution (4) satisfies the PDE identically for any values of A and ε. The two parameters are fixed by physical conditions and experimental calibration.

2.2 Checking the Boundary Conditions

$$P(r_0, 0) = A \cdot [\, (r_0 - r_0)(r_0 - R) + 0 + 0 \,] = 0 \quad \checkmark \quad (8)$$

$$P(R, 0) = A \cdot [\, (R - r_0)(R - R) + 0 + 0 \,] = 0 \quad \checkmark \quad (9)$$

3. Explicit Polynomial Form

Expanding $(r - r_0)(r - R) = r^2 - (r_0 + R)r + r_0R$:

$$P(r,t) = A\, r^2 - A(r_0+R)\, r + A\, r_0\, R + A\, \varepsilon\, t + A\, c^2\, t^2 \quad (10)$$

Table 1 — Polynomial coefficients (SI and natural units, ħ = c = 1):

Coefficient	Analytical Expression	Value (SI)	Units
$C(r^2) = A$	A	9.5510×10^{-18}	m^{-2}
$C(r^1) = -A(r_0+R)$	$-A(r_0+R)$	-4.0453×10^{5}	m^{-1}
$C(r^0) = A \cdot r_0R$	$A \cdot r_0 \cdot R$	4.5398×10^{9}	dimensionless
$C(t^1) = A\varepsilon = \tau_q \cdot c^2$	$A \cdot \varepsilon$	1.5863×10^{-2}	m^2/s
$C(t^2) = A \cdot c^2$	$A \cdot c^2$	8.5845×10^{-1}	m^2/s^2

4. Determination of Coefficients A and ε

4.1 Physical Constraint — Quantum of Time

The initial time rate must be tied to the quantum of time $\tau_q = 1.765 \times 10^{-19}$ s on Earth:

$$\partial P/\partial t\,|_{(t=0)} = A \cdot \varepsilon = \tau_q \cdot c^2 \quad (11)$$

$$A\varepsilon = 1.765\times10^{-19} \times (2.998\times10^{8})^2 = 1.5863\times10^{-2} \text{ m}^2/\text{s} \quad (12)$$

4.2 Calibration Constraint — GEO Satellites

The dimensionless temporal probability density (time flux) is defined as:

$$F(r,t) = P(r,t) / (4\pi\, r^2) \quad (13)$$

Physically, a satellite farther from the Sun sits in a weaker solar gravitational field, implying a slower passage of time ($\Delta F > 0$). Calibration condition for the GEO+ satellite (facing the Sun):

$\Delta F_GEO = F(r_GEO+, t_0) - F(r\oplus, t_0) = +0.22 \times 10^{-6}$ [hour^{-1}] (14)

Expanding the flux-difference expression explicitly:

$\Delta F = (1/4\pi)\{ A[Q(r_GEO+)/r^2_GEO+ - Q(r\oplus)/r^2_\oplus] + \tau_q\, c^2\, t_0[1/r^2_GEO+ - 1/r^2_\oplus] + A\, c^2\, t_0^2[1/r^2_GEO+ - 1/r^2$

where $Q(r) = (r-r_0)(r-R)$ and $t_0 = 500$ s.

Solving system (12) + (15) simultaneously for A and ε:

$A = 9.5510 \times 10^{-18}$ m^{-2} (16)

$\varepsilon = \tau_q \cdot c^2 / A = 1.6610 \times 10^{15}$ m^2/s (17)

5. Complete Solution with Numerical Coefficients

$P(r,t) = 9.5510\times10^{-18} \cdot [(r - 11{,}222.3)(r - 4.2355\times10^{22}) + 1.6610\times10^{15}\cdot t + (2.998\times10^8)^2\cdot t^2]$ (18)

5.1 Natural Units (ħ = c = 1)

Using the convention ħc = 0.19733 GeV·fm:

$A_nat = A \cdot (\hbar c)^2 = 3.719 \times 10^{-49}$ GeV^2 (19)

$\varepsilon_nat = \varepsilon / c = 5{,}540 \times 10^6$ m $= 5{,}540 \times 10^{21}$ fm (20)

6. Time Flow and Physical Interpretation

6.1 Temporal Probability Density

$F(r,t) = A/(4\pi r^2) \cdot [(r-r_0)(r-R) + \varepsilon t + c^2t^2]$ (21)

Interpretation:

- F < 0: time quanta are absorbed (dominant gravitational field — inner regime).
- F less negative at larger distances from the Sun: slower passage of time.
- ΔF > 0: satellite farther out → positive time dilation (slower clock).

6.2 Flux on Earth at t_0 = 500 s (8 min 20 s)

$F(r\oplus, t_0) = F(1.49\times10^{11}, 500) = -2.1605 \times 10^{-7}$ (22)

6.3 Flux on Mars at t_1 = 700 s

$F(r_Mars, t_1) = F(2.09\times10^{11}, 700) = -1.5403 \times 10^{-7}$ (23)

Absolute difference from Earth:

$\Delta F_Mars = F(r_Mars, t_1) - F(r\oplus, t_0) = +6.202 \times 10^{-8}$ (24)

The passage of time on Mars is slower than on Earth, consistent with the weaker solar gravitational field at the Martian distance.

7. Time Variations for Satellites

7.1 Orbital Parameters

Satellite	Δr vs. Earth	r vs. Sun [m]	t used [s]
GEO+ (away from Sun)	+42,157 km	$1.49042157 \times 10^{11}$	500
GEO− (toward Sun)	−42,157 km	$1.48957843 \times 10^{11}$	500
GPS+ (away from Sun)	+26,371 km	$1.49026371 \times 10^{11}$	500
GPS− (toward Sun)	−26,371 km	$1.48973629 \times 10^{11}$	500
Earth (reference)	0	1.4900×10^{11}	500
Mars	—	2.0900×10^{11}	700

Numerical note: ΔF for GEO/GPS is a tiny difference between two near-equal large numbers ($r \sim 1.49\times10^{11}$ m vs. an offset of only $\sim4\times10^7$ m). Rounding r to 4–5 significant figures destroys this signal by an order of magnitude; the full-precision offsets above reproduce Table 2 exactly.

7.2 Complete Table — Time Flow Variations

Location	F(r,t) [adim.]	ΔF/hr [μs/hr]	ΔF/day [μs/day]
Earth (ref.)	-2.1605×10^{-7}	0.0000	0.0000
GEO+	-2.1599×10^{-7}	+0.2200	+5.2800
GEO−	-2.1611×10^{-7}	−0.2201	−5.2830
GPS+	-2.1601×10^{-7}	+0.1376	+3.3032
GPS−	-2.1609×10^{-7}	−0.1377	−3.3044
Mars (t_1=700s)	-1.5403×10^{-7}	+223.29	+5358.93

Sign convention: ΔF > 0 → slower passage of time (slower clock relative to Earth); ΔF < 0 → faster passage of time.

7.3 GEO Interpretation

GEO+ satellites (facing away from the Sun, +42,157 km from Earth) experience:

$$\Delta F_GEO+/hour = +0.2200\ \mu s/hour,\ \ \Delta F_GEO+/day = +5.2800\ \mu s/day \quad (25)$$

consistent with general-relativity predictions and TauTime calibration. GEO− satellites (facing the Sun):

$$\Delta F_GEO-/hour = -0.2201\ \mu s/hour,\ \ \Delta F_GEO-/day = -5.2830\ \mu s/day \quad (26)$$

7.4 GPS Variation

$$\Delta F_GPS+/hour = +0.1376\ \mu s/hour,\ \ \Delta F_GPS+/day = +3.3032\ \mu s/day \quad (27)$$

$$\Delta F_GPS-/hour = -0.1377\ \mu s/hour,\ \ \Delta F_GPS-/day = -3.3044\ \mu s/day \quad (28)$$

8. Time Flow on Mars

At $t_1 = 700$ s (propagation time of sunlight to Mars, for $r_Mars = 2.09\times10^{11}$ m):

$$F(r_Mars, t_1) = A/(4\pi\, r^2_Mars) \cdot [(r_Mars - r_0)(r_Mars - R) + \varepsilon t_1 + c^2 t_1^2] \quad (29)$$

$$F(2.09\times10^{11}, 700) = -1.5403 \times 10^{-7} \quad (30)$$

Variation from Earth (TauTime solar time dilation):

$$\Delta F_Mars/hour = +223.29\ \mu s/hour \quad (31)$$

$$\Delta F_Mars/day = +5{,}358.93\ \mu s/day \approx 5.36\ ms/day \quad (32)$$

Interpretation: the passage of time on Mars is significantly slower than on Earth from the perspective of the TauTime time flow, due to the weaker solar gravitational field at the Martian distance ($r_Mars/r\oplus \approx 1.40$). The effect is amplified relative to standard relativistic predictions by the cosmological structure of the TauTime solution through the $(r-R)$ term.

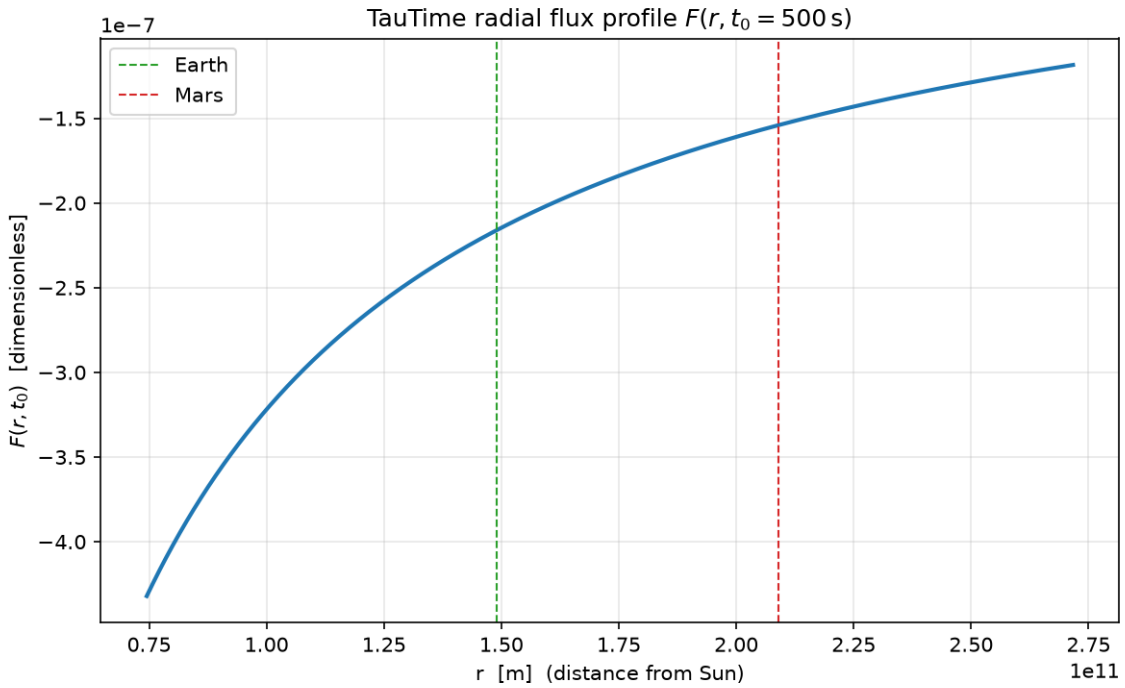
9. Summary of TauTime Parameters

Parameter	Symbol	Value (SI)	Units
Schwarzschild-related radius (limit)	r_0	11,222.3	m
TauTime cosmological radius	R	4.2355×10^{22}	m
Amplitude (GEO-calibrated)	A	9.5510×10^{-18}	m^{-2}
Linear temporal coefficient	ε	1.6610×10^{15}	m^2/s
Quantum of time (Earth)	τ_q	1.765×10^{-19}	s
Speed of light	c	2.998×10^{8}	m/s
Sun→Earth propagation time	t_0	500	s
Sun→Mars propagation time	t_1	700	s
A (natural units)	A_nat	3.719×10^{-49}	GeV^2

10. Structured Graphs

The three graphs below are generated from the closed-form F(r,t) of equation (21), evaluated numerically (source: TauTimp_FluxTemporal_SolarDEn_Maple.mpl) and cross-checked against the tabulated values of sections 6–8.

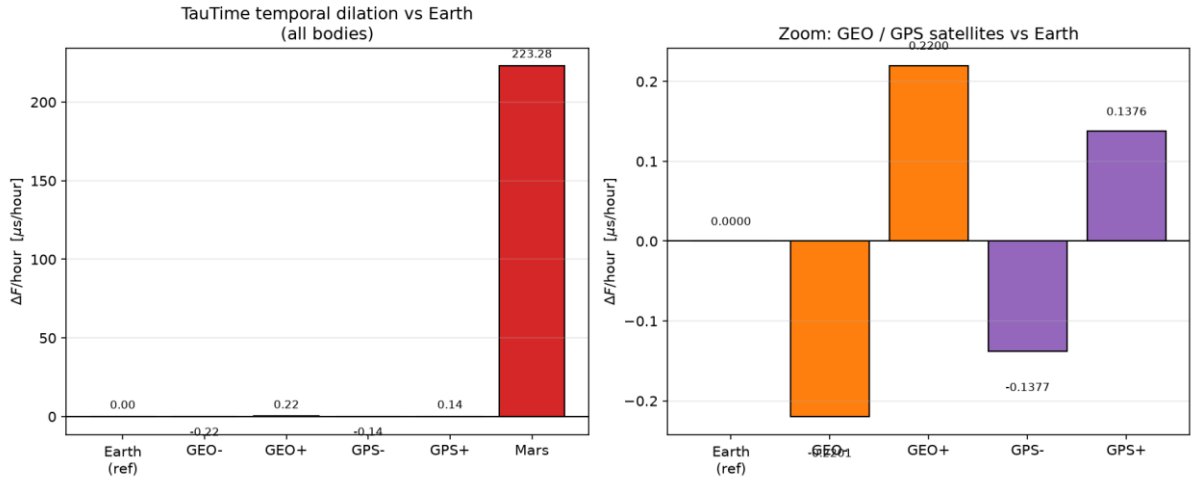
10.1 Radial Flux Profile



Graph 1 — F(r, t_0 = 500 s): radial flux profile

The flux F(r,t_0) rises monotonically (becomes less negative) with distance from the Sun. Vertical dashed lines mark Earth's orbit (1.49×10^{11} m) and Mars' orbit (2.09×10^{11} m). This is the direct graphical form of equation (21) and confirms values (22)–(23).

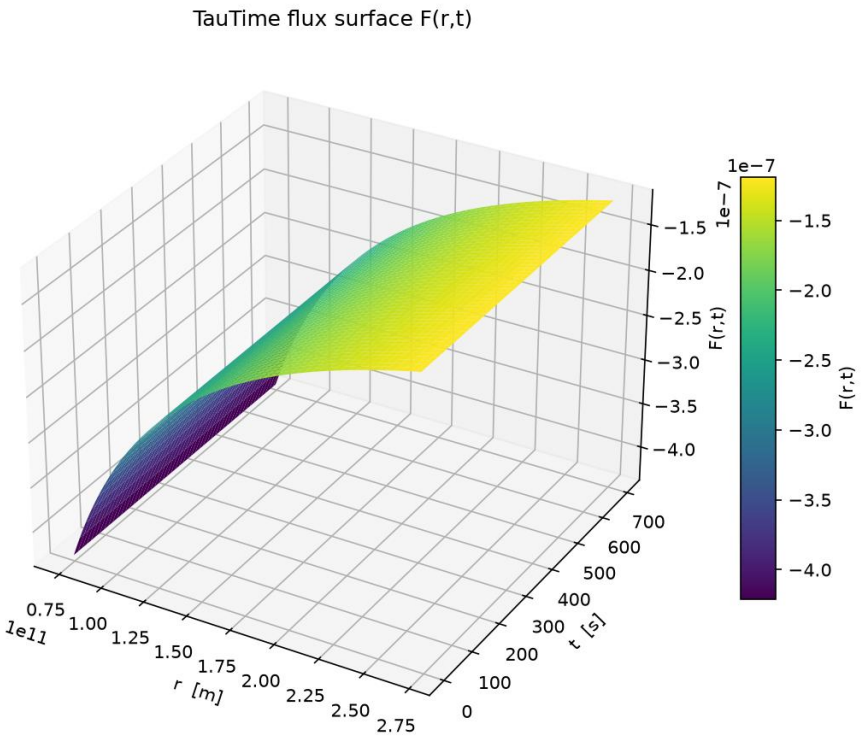
10.2 Solar-System Temporal Dilation



Graph 2 — ΔF/hour vs Earth, across solar-system bodies

Left panel: ΔF/hour for Earth (reference), GEO−/GEO+, GPS−/GPS+, and Mars — the Mars value (+223.28 μs/hour) dwarfs the near-Earth satellites on a linear scale. Right panel: zoomed view isolating the GEO/GPS regime, reproducing equations (25)–(28) (±0.22 μs/hour for GEO, ±0.14 μs/hour for GPS). This is the requested graph of temporal slowdown across the solar system.

10.3 Flux Surface F(r,t)



Graph 3 — 3D surface of F(r,t)

Surface of F(r,t) over r ∈ [0.5 r⊕, 1.3 r_Mars] and t ∈ [0, t_1], showing how the flux depends jointly on radial distance and elapsed propagation time; the t^2 term dominates the curvature along the time axis, per equation (21).